

Thruster integration - design brief

AI BRAKING / AB-ACT / C0 / 2026-09-07

A mounting cradle and simplified thruster envelope support early integration. The motor, pump, piston and seal internals are deliberately not released as manufacturable designs.

TARGETS (not validated)

Illustrative stroke: 60 mm. Envelope assumption

Illustrative release force: 2 kN. RFQ starting point, not supplier selection

Body envelope: Ø120 × 240 mm. Placeholder

Electrical supply / duty: To be specified. Requirements only

DESIGN DECISIONS

Buy an industrial thruster for the first brake prototypes.

Keep mounting clevises replaceable to adapt to the selected supplier drawing.

Separate thrust, stroke and time requirements; one cannot substitute for another.

CANDIDATE MATERIALS

Cradle: S355J2+N candidate fabricated plate.

Thruster body and rod: supplier-controlled; CAD volumes are reserved space only.

Assembly and engineering gates

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ASSEMBLY INTENT

- 1. Select supplier using force-stroke-time-temperature data.
- 2. Replace envelope with supplier STEP and approved pin dimensions.
- 3. Fit cradle and retained pivots; inspect alignment across full stroke.

OPEN DESIGN RISKS

- 1. Force requirement cannot be finalized until the brake linkage is solved.
- 2. No internal hydraulic/electrical design, pressure rating or environmental certification.

VALIDATION REQUIRED

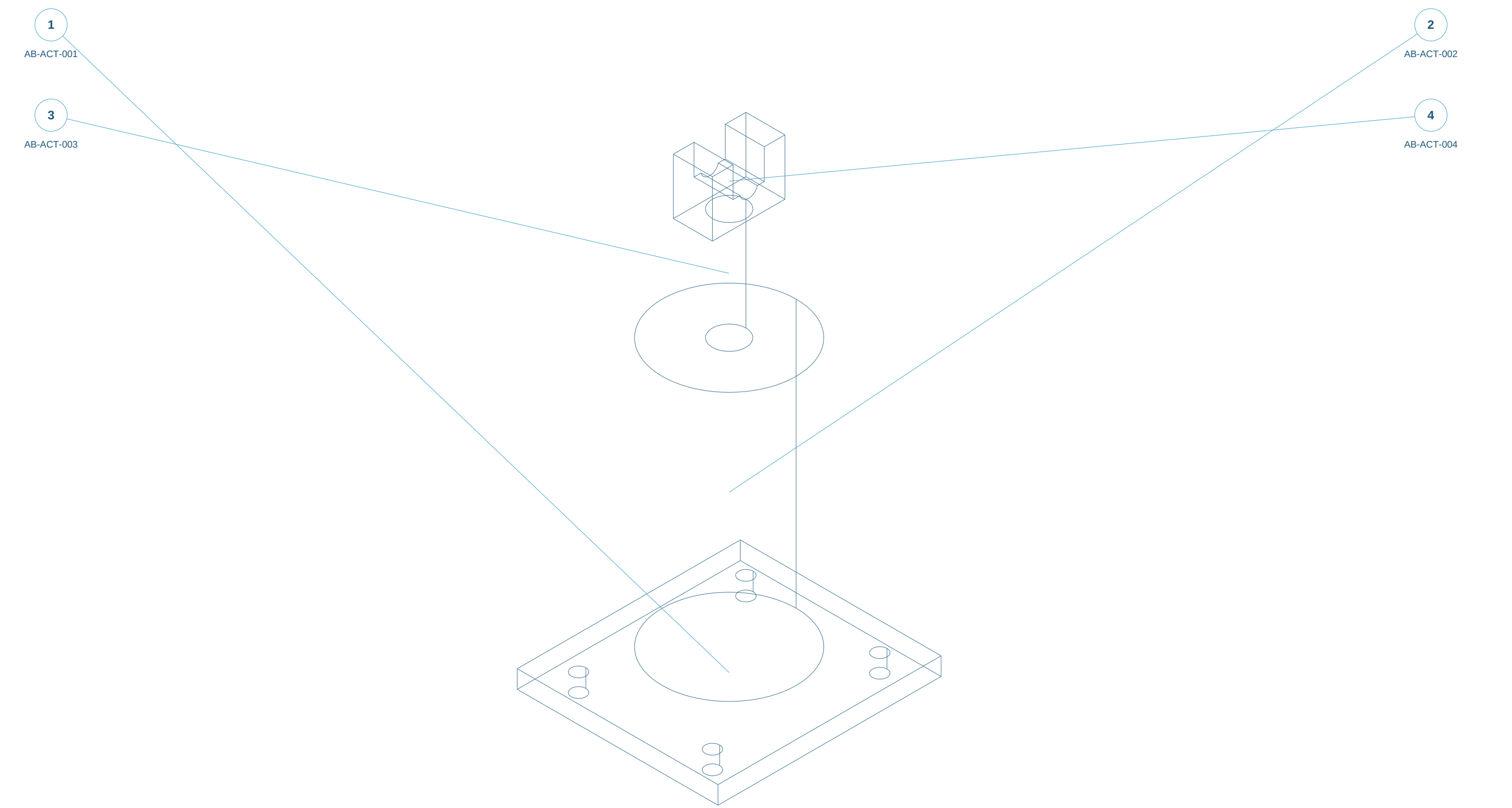
- 1. Supplier documentation review and incoming force/stroke/time tests.
- 2. Side-load, mounting fatigue, seal leakage and temperature tests.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

- 1. Specify motor voltage/frequency, duty cycle, insulation, cable entry, ingress, thermal protection and end-state sensing.

Assembly reference / numbered components

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Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

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AB-ACT-001 | Integration base | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-ACT-002 | Thruster body envelope | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy selected electrohydraulic thruster

AB-ACT-003 | Rod travel envelope | interface-envelope

Supplier-controlled; illustrative reserved envelope; Supplier model required

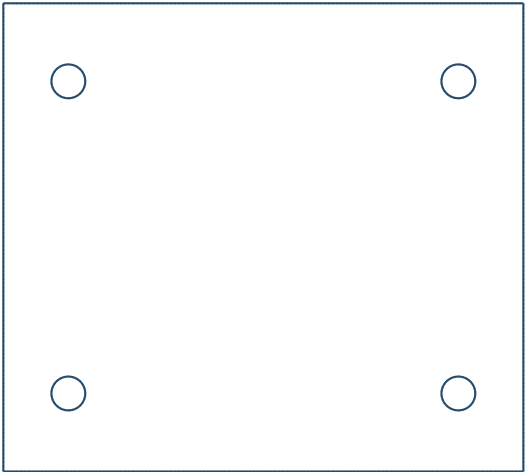
AB-ACT-004 | Top clevis envelope | interface-envelope

Supplier-controlled; illustrative reserved envelope; Supplier interface

AB-ACT-001 / Integration base

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TOP X-Y



FRONT X-Z



S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 200.00 / 180.00 / 16.00 mm

200 x 180 x 16; 4 x diameter 13 at (+/-75,+/-60).

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

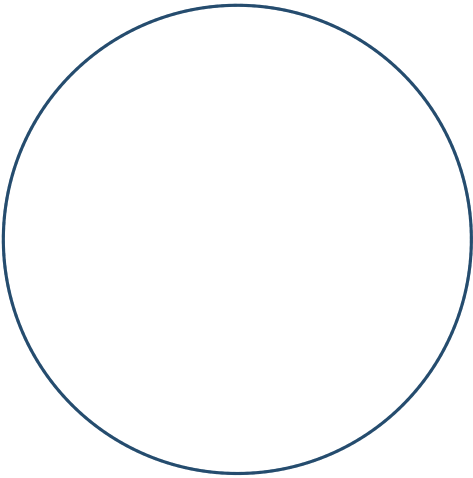
SIDE Y-Z



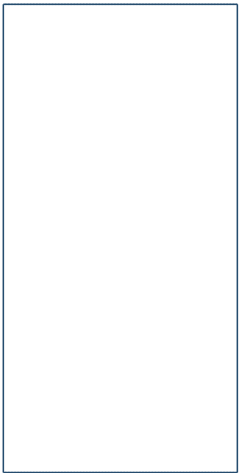
AB-ACT-002 / Thruster body envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Buy selected electrohydraulic thruster

Overall X/Y/Z bounds: 120.00 / 120.00 / 240.00 mm

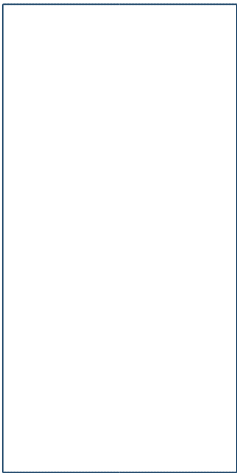
All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

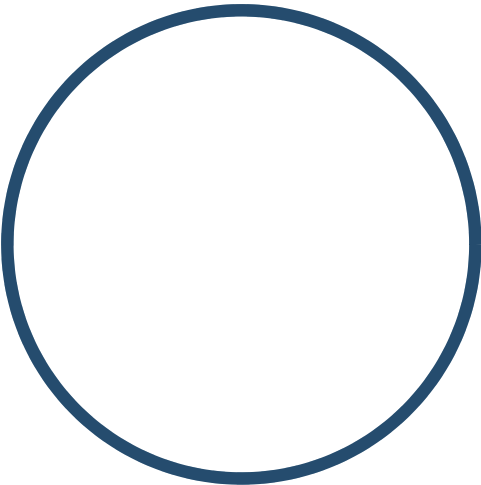
SIDE Y-Z



AB-ACT-003 / Rod travel envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Supplier model required

Overall X/Y/Z bounds: 30.00 / 30.00 / 100.00 mm

Includes 60 mm assumed stroke; no seal or bearing design.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

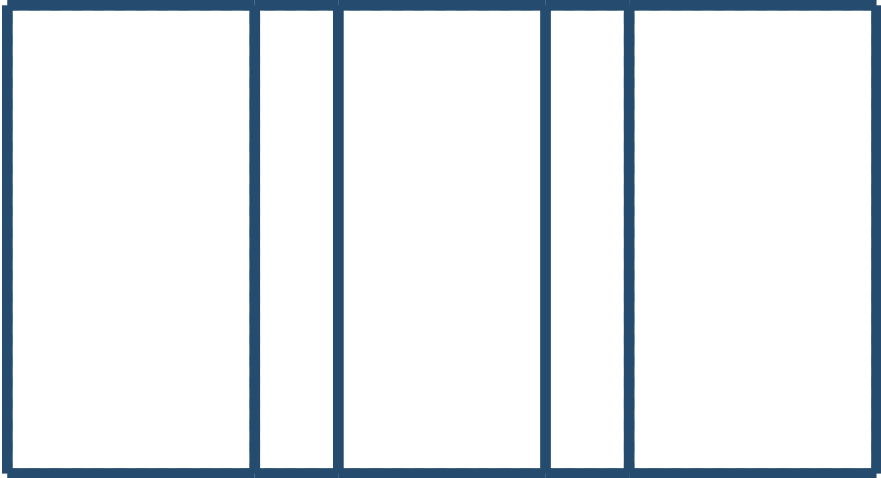
SIDE Y-Z



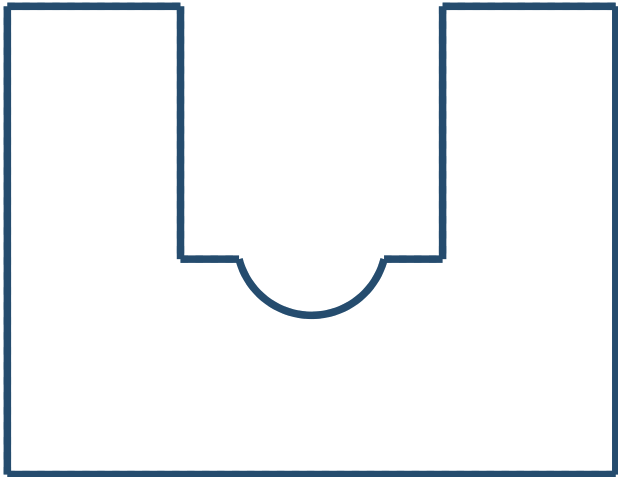
AB-ACT-004 / Top clevis envelope

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TOP X-Y



FRONT X-Z



Supplier-controlled; illustrative reserved envelope

Supplier interface

Overall X/Y/Z bounds: 65.00 / 35.00 / 50.00 mm

All dimensions nominal. Fits, GD&T and edge details awaiting engineer release.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

SIDE Y-Z

